```
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
      name:  <unnamed>
       log:  C:\Evaluation course\RCT\
RCT_Final_Complete_Results_Baseline_Endline_Results.log
  log type:  text
 opened on:  16 Sep 2026, 00:04:11

.
.  *IMPACT EVALUATION USING RCT EXPERIMENTAL METHOD
.  *PIPED WATER CONNECTION INTERVENTION
.  * Author: Meshack Omondi
.  ******************************************************************************
.  ******************************************************************************
.
.  ******************************************************************************
.  *1. DATA IMPORT AND INITIAL CHECKS
.  ******************************************************************************
.  import excel using "Randomized_Assignment_Final.xlsx", sheet("F_Baseline_data")
firstrow clear

.  describe

Contains data
  obs:         1,032
 vars:            10
 size:        54,696
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
HouseholdID     str6    %9s                   Household ID
Village         str12   %12s                  Village
Household_Size  byte    %10.0g                Household_Size
Monthly_House~e long    %10.0g                Monthly_Household_Income
Current_Water~e str9    %9s                   Current_Water_Source
Time_Spent_Co~r int     %10.0g                Time_Spent_Collecting_water
Distance_to_H~y double  %10.0g                Distance_to_Health_facility
Monthly_Healt~e int     %10.0g                Monthly_Health_Expenditure
RandomNo        double  %10.0g                Random No.
treatment       byte    %10.0g                treatment
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
Sorted by:
     Note: Dataset has changed since last saved.

.  summarize
```

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
 HouseholdID |          0
     Village |          0
Household_~e |      1,032    5.409884    2.204312          1         10
Monthly_Ho~e |      1,032    41421.32    23647.03       6800     150000
Current_Wa~e |          0
-------------+---------------------------------------------------------
Time_Spent~r |      1,032    53.65698    25.98333         11        120
Distance_t~y |      1,032    6.759109    3.800719         .2         15
Monthly_He~e |      1,032    5983.721    3293.026          0      16300
    RandomNo |      1,032     .503525    .2972335   .0005103   .9998322
   treatment |      1,032          .5    .5002424          0          1

.  codebook

-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
HouseholdID
                                                                  Household ID
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                  type:  string (str6)

         unique values:  1,032                    missing "":  0/1,032

              examples:  "HH173"
                         "HH379"
                         "HH586"
                         "HH792"

-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
Village
                                                                      Village
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                  type:  string (str12)

         unique values:  4                        missing "":  0/1,032

            tabulation:  Freq.  Value
                           243  "Kiserian"
                           227  "Namanga"
                           292  "Oloitoktok A"
                           270  "Oloitoktok B"

               warning:  variable has embedded blanks
```

```
-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Household_Size
                                                              Household_Size
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  numeric (byte)

                 range:  [1,10]                       units:  1
          unique values:  10                       missing .:  0/1,032

                  mean:   5.40988
             std. dev:   2.20431

            percentiles:        10%       25%       50%       75%       90%
                                  3         4         5         7         8

-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Monthly_Household_Income
                                                 Monthly_Household_Income
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  numeric (long)

                 range:  [6800,150000]                units:  100
          unique values:  542                      missing .:  0/1,032

                  mean:   41421.3
             std. dev:     23647

            percentiles:        10%       25%       50%       75%       90%
                             17900     25200     36000     51900     70400

-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Current_Water_Source
                                                     Current_Water_Source
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  string (str9)

          unique values:  4                        missing "":  0/1,032

             tabulation:  Freq.  Value
                            235  "Borehole"
```

```
                          293  "Rainwater"
                          287  "River"
                          217  "Vendor"

-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Time_Spent_Collecting_water
                                                Time_Spent_Collecting_water
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  numeric (int)

                 range:  [11,120]                     units:  1
         unique values:  110                      missing .:  0/1,032

                  mean:    53.657
             std. dev:   25.9833

           percentiles:        10%       25%       50%       75%       90%
                                22        34        51        67        95

-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Distance_to_Health_facility
                                             Distance_to_Health_facility
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  numeric (double)

                 range:  [.2,15]                      units:  .1
         unique values:  149                      missing .:  0/1,032

                  mean:   6.75911
             std. dev:   3.80072

           percentiles:        10%       25%       50%       75%       90%
                               1.7       3.5       6.6       9.7        12

-------------------------------------------------------------------------------
-------------------------------------------------------------------------
Monthly_Health_Expenditure
                                              Monthly_Health_Expenditure
-------------------------------------------------------------------------------
-------------------------------------------------------------------------

                  type:  numeric (int)

                 range:  [0,16300]                    units:  100
```

```
         unique values:  144                      missing .:  0/1,032

                  mean:   5983.72
             std. dev:   3293.03

           percentiles:        10%       25%       50%       75%       90%
                              1600      3700      5900      8300     10300
```

RandomNo

Random No.

```
                 type:  numeric (double)

                range:  [.0005103,.99983217]         units:  1.000e-12
         unique values:  1,032                    missing .:  0/1,032

                  mean:   .503525
             std. dev:   .297234

           percentiles:        10%       25%       50%       75%       90%
                            .091998    .23499   .514361   .770088   .916655
```

---

treatment

treatment

```
                 type:  numeric (byte)

                range:  [0,1]                        units:  1
         unique values:  2                        missing .:  0/1,032

            tabulation:  Freq.  Value
                           516  0
                           516  1

.  tabulate treatment

  treatment |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        516       50.00       50.00
          1 |        516       50.00      100.00
------------+-----------------------------------
      Total |      1,032      100.00
```

```
. 
. ******************************************************************************
. *2. BASELINE Descriptive statistics
. ******************************************************************************
. * Overall descriptive statistics
. summarize Household_Size Monthly_Household_Income ///
>     Time_Spent_Collecting_water Distance_to_Health_facility ///
>     Monthly_Health_Expenditure

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
Household_~e |      1,032    5.409884    2.204312          1         10
Monthly_Ho~e |      1,032    41421.32    23647.03       6800     150000
Time_Spent~r |      1,032    53.65698    25.98333         11        120
Distance_t~y |      1,032    6.759109    3.800719         .2         15
Monthly_He~e |      1,032    5983.721    3293.026          0      16300

. 
. * Descriptive statistics by treatment group
. tabstat Household_Size Monthly_Household_Income ///
>     Time_Spent_Collecting_water Distance_to_Health_facility ///
>     Monthly_Health_Expenditure, ///
>     by(treatment) statistics(n mean sd min max)

Summary statistics: N, mean, sd, min, max
  by categories of: treatment (treatment)

treatment |  Househ~e  Month~me  Time_S~r  Distan~y  Month~re
----------+---------------------------------------------------
        0 |       516       516       516       516       516
          |  5.315891  41494.96  53.02907  6.767636  5822.481
          |  2.231073  21783.03  26.02115  3.880819  3219.473
          |         1      8000        11        .2         0
          |        10    150000       120        15     14500
----------+---------------------------------------------------
        1 |       516       516       516       516       516
          |  5.503876  41347.67  54.28488  6.750581  6144.961
          |  2.175322  25395.64  25.95549  3.722644  3360.356
          |         1      6800        11        .2         0
          |        10    150000       120        15     16300
----------+---------------------------------------------------
    Total |      1032      1032      1032      1032      1032
          |  5.409884  41421.32  53.65698  6.759109  5983.721
          |  2.204312  23647.03  25.98333  3.800719  3293.026
          |         1      6800        11        .2         0
          |        10    150000       120        15     16300
------------------------------------------------------------

. 
```

```
. * Baseline water source distribution
. tabulate Current_Water_Source

Current_Wat |
  er_Source |      Freq.     Percent        Cum.
------------+-----------------------------------
   Borehole |        235       22.77       22.77
  Rainwater |        293       28.39       51.16
      River |        287       27.81       78.97
     Vendor |        217       21.03      100.00
------------+-----------------------------------
      Total |      1,032      100.00

.
. * Baseline water source by treatment group
. tabulate Current_Water_Source treatment, row

+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+

Current_Wa |       treatment
ter_Source |         0          1 |     Total
-----------+----------------------+----------
  Borehole |       108        127 |       235
           |     45.96      54.04 |    100.00
-----------+----------------------+----------
 Rainwater |       144        149 |       293
           |     49.15      50.85 |    100.00
-----------+----------------------+----------
     River |       143        144 |       287
           |     49.83      50.17 |    100.00
-----------+----------------------+----------
    Vendor |       121         96 |       217
           |     55.76      44.24 |    100.00
-----------+----------------------+----------
     Total |       516        516 |     1,032
           |     50.00      50.00 |    100.00


.
. * Number of households by village
. tabulate Village

     Village |      Freq.     Percent        Cum.
-------------+-----------------------------------
    Kiserian |        243       23.55       23.55
```

```
     Namanga |        227       22.00       45.54
Oloitoktok A |        292       28.29       73.84
Oloitoktok B |        270       26.16      100.00
-------------+-----------------------------------
       Total |      1,032      100.00
```

```
. 
. * Treatment allocation by village
. tabulate Village treatment, row
```

```
+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+
```

```
             |       treatment
     Village |         0          1 |     Total
-------------+----------------------+----------
    Kiserian |       119        124 |       243
             |     48.97      51.03 |    100.00
-------------+----------------------+----------
     Namanga |       105        122 |       227
             |     46.26      53.74 |    100.00
-------------+----------------------+----------
Oloitoktok A |       152        140 |       292
             |     52.05      47.95 |    100.00
-------------+----------------------+----------
Oloitoktok B |       140        130 |       270
             |     51.85      48.15 |    100.00
-------------+----------------------+----------
       Total |       516        516 |     1,032
             |     50.00      50.00 |    100.00
```

```
. *============================================================*
. * 3. BASELINE BALANCE ASSESSMENT
. *============================================================*
. 
. * Household size
. ttest Household_Size, by(treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     516    5.315891    .0982175    2.231073    5.122935    5.508848
       1 |     516    5.503876    .0957632    2.175322    5.315741    5.692011
---------+--------------------------------------------------------------------
```

```
combined |   1,032    5.409884    .0686172    2.204312    5.275238    5.544529
---------+--------------------------------------------------------------------
    diff |           -.1879845    .1371761               -.4571609     .081192
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -1.3704
Ho: diff = 0                                     degrees of freedom =     1030

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0854         Pr(|T| > |t|) = 0.1709          Pr(T > t) = 0.9146

.
. * Monthly household income
. ttest Monthly_Household_Income, by(treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     516    41494.96    958.9446    21783.03    39611.04    43378.89
       1 |     516    41347.67    1117.981    25395.64    39151.31    43544.04
---------+--------------------------------------------------------------------
combined |   1,032    41421.32       736.1    23647.03    39976.89    42865.74
---------+--------------------------------------------------------------------
    diff |            147.2868    1472.907               -2742.955    3037.529
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =   0.1000
Ho: diff = 0                                     degrees of freedom =     1030

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.5398         Pr(|T| > |t|) = 0.9204          Pr(T > t) = 0.4602

.
. * Time spent collecting water
. ttest Time_Spent_Collecting_water, by(treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     516    53.02907    1.145517    26.02115    50.77861    55.27953
       1 |     516    54.28488    1.142627    25.95549     52.0401    56.52967
---------+--------------------------------------------------------------------
combined |   1,032    53.65698    .8088257    25.98333    52.06984    55.24411
---------+--------------------------------------------------------------------
    diff |           -1.255814    1.617963               -4.430695    1.919067
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -0.7762
Ho: diff = 0                                     degrees of freedom =     1030

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
```

```
 Pr(T < t) = 0.2189         Pr(|T| > |t|) = 0.4378          Pr(T > t) = 0.7811

.
. * Distance to health facility
. ttest Distance_to_Health_facility, by(treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     516    6.767636    .1708435    3.880819       6.432    7.103272
       1 |     516    6.750581    .1638803    3.722644    6.428625    7.072537
---------+--------------------------------------------------------------------
combined |   1,032    6.759109    .1183112    3.800719     6.52695    6.991267
---------+--------------------------------------------------------------------
    diff |            .0170543    .2367367               -.447487    .4815955
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =   0.0720
Ho: diff = 0                                     degrees of freedom =     1030

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.5287         Pr(|T| > |t|) = 0.9426          Pr(T > t) = 0.4713

.
. * Monthly household health expenditure
. ttest Monthly_Health_Expenditure, by(treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     516    5822.481    141.7294    3219.473    5544.042     6100.92
       1 |     516    6144.961    147.9314    3360.356    5854.338    6435.585
---------+--------------------------------------------------------------------
combined |   1,032    5983.721    102.5074    3293.026    5782.574    6184.868
---------+--------------------------------------------------------------------
    diff |           -322.4806    204.8681               -724.4871    79.52585
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -1.5741
Ho: diff = 0                                     degrees of freedom =     1030

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0579         Pr(|T| > |t|) = 0.1158          Pr(T > t) = 0.9421

.
. * Baseline water source
. tabulate Current_Water_Source treatment, chi2 row

+----------------+
| Key            |
```

```
|----------------|
|   frequency    |
| row percentage |
+----------------+

Current_Wa |       treatment
ter_Source |         0          1 |     Total
-----------+----------------------+----------
  Borehole |       108        127 |       235
           |     45.96      54.04 |    100.00
-----------+----------------------+----------
 Rainwater |       144        149 |       293
           |     49.15      50.85 |    100.00
-----------+----------------------+----------
     River |       143        144 |       287
           |     49.83      50.17 |    100.00
-----------+----------------------+----------
    Vendor |       121         96 |       217
           |     55.76      44.24 |    100.00
-----------+----------------------+----------
     Total |       516        516 |     1,032
           |     50.00      50.00 |    100.00

          Pearson chi2(3) =   4.5052   Pr = 0.212

.
end of do-file
```